Financial Profiles of Italian SMEs

Clustering results for the IFCS 2026 Data Challenge

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FY2023 financial statements of 13,956 SMEs

Clustering Question

Goal

- Identify financial profiles of Italian SMEs.
- Explain how these profiles differ economically.
- Examine where fragile profiles are concentrated.

Dataset

Firms	13,956
Distressed firms	1,505
Distress rate	10.78%
Provinces	108
Sectors	306

Key choice: clustering uses financial variables only. Geography, sector, ATECO, and the distress target are used after clustering to interpret the groups.

Financial Variables Used

Financial signal

- Size: revenue and employees.
- Profitability: net and operating margins.
- Cash generation: operating cash flow and OCF margin.
- Debt pressure: financial expenses and Alert Index.
- Productivity: revenue and income per employee.

Preprocessing

- Signed log transforms for skewed monetary variables.
- Winsorization at 0.5th and 99.5th percentiles.
- Median imputation.
- Robust scaling.
- K-means tested for $k = 2, \ldots, 8$.

Chosen solution

k = 5: interpretable financial health ladder

Five SME Financial Profiles

Cluster	Firms	Distress	Net margin	OCF margin
A Resilient high-profit cash generators	2,085	0.43%	15.6%	19.7%
B Healthy profitable core SMEs	6,043	2.75%	4.5%	7.4%
C Thin-margin vulnerable operators	4,745	15.68%	0.7%	3.2%
D Debt-burdened break-even firms	174	31.61%	-1.8%	1.5%
E Loss-making cash-flow distressed firms	909	58.42%	-10.2%	-6.1%

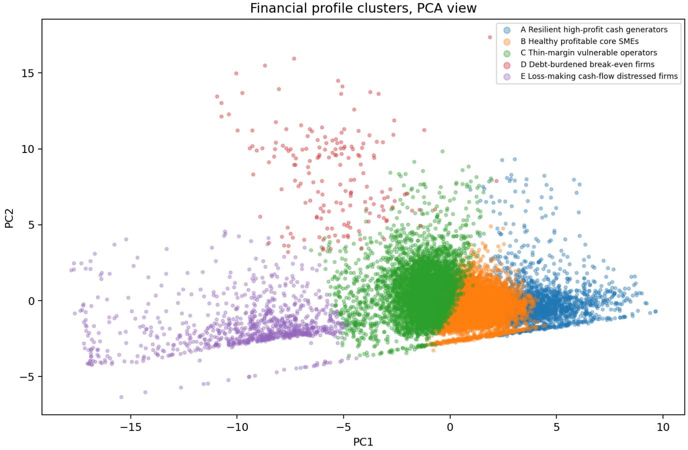
Low-risk side

- Clusters A and B generate profits and cash.
- Distress is below 3% in both profiles.

High-risk side

- Cluster C is the large vulnerable middle.
- Cluster D is small but debt-burdened.
- Cluster E combines losses and negative cash flow.

PCA View of the Financial Profiles



PCA components

- **PC1 = profitability and cash generation**
- Explains 52.7% of scaled variance.
- Positive: operating income, cash flow, net margin, operating margin.
- Negative: high financial-expense ratios and negative-income flags.
- **PC2 = debt-service pressure**
- Explains 14.4% of scaled variance.
- Positive: financial expenses relative to sales and operating income.

Centroids: A/B are on the healthy PC1 side; C is weaker but central; D is high on PC2; E is strongly negative on PC1.

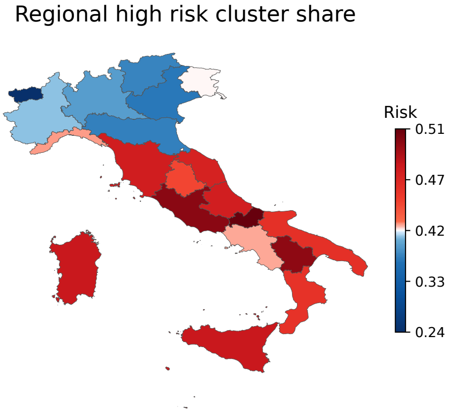
Sector Mix Explains Why Some Profiles Are More Exposed

Top over-represented sectors per cluster, by lift vs. each sector's overall share of the dataset.

A Cash generators	B Healthy core	C Thin margins	D Debt-burdened	E Loss-making
Insurance & pension funds **2.8x**	Non-specialized wholesale trade **1.5x**	Specialized food & beverage retail **1.7x**	Rubber products manufacturing **4.7x**	R&D – natural sciences & engineering **5.0x**
Professional & scientific services **2.1x**	Wholesale – ICT equipment **1.4x**	Restaurants & mobile food service **1.5x**	Printing services **4.3x**	Sports activities **4.2x**
Human health services **2.0x**	Wholesale – specialized products **1.4x**	Shipbuilding **1.5x**	Beverage manufacturing **3.8x**	Real estate renting & operating **4.0x**

Note: sector is not used to form the clusters; this is a post-clustering lift analysis using sectors with at least 30 firms.

Geography of Fragile Profiles



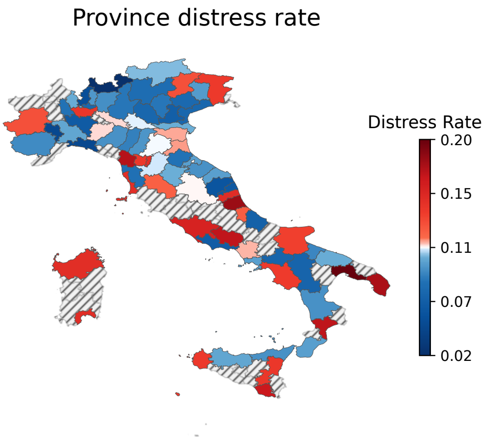
High-risk profiles

- C: thin-margin vulnerable
- D: debt-burdened break-even
- E: loss-making cash-flow distressed

Region	High-risk share
Molise	51.43%
Lazio	50.17%
Basilicata	50.00%
Sicilia	48.01%
Sardegna	47.96%

This is a concentration pattern, not a causal claim about location.

Province-Level Distress Check



Target-side view

- This map shows observed distress rates by province.
- It is not used to form the clusters.
- It checks whether regional cluster-risk patterns line up with local distress pockets.

Main Takeaways

1. The clusters are economically meaningful

The k=5 solution separates SMEs by profitability, cash generation, productivity, and debt-service pressure.

2. Distress follows the financial health ladder

Distress rises from 0.43% in the strongest cluster to 58.42% in the loss-making cash-flow cluster.

3. Geography and sector enrich the interpretation

Fragile profiles are not uniformly distributed: some regions and activities contain a larger share of thin-margin, debt-burdened, or loss-making firms.

The clustering turns financial statements into an interpretable risk map.